

Jurisdiction–Aware Credit Decision Governance Monitor

Project presentation | Revolut | US + EU | Option C analytical design with quantitative prototype

Core thesis: shared governance schema $\neq$ shared sensitive–data pipeline

Tool produces evidence + escalation, not credit decisions

What I built

A RegTech governance monitor for AI–assisted credit decisioning, using synthetic data and a rule configuration layer to produce jurisdiction–specific evidence.

Why it matters

US and EU credit AI rules encode different choices: individual adverse–action explanation versus high–risk AI lifecycle governance.

What it outputs

US/EU evidence packages with risk signal, owner, required action, deadline, escalation level and audit reference.

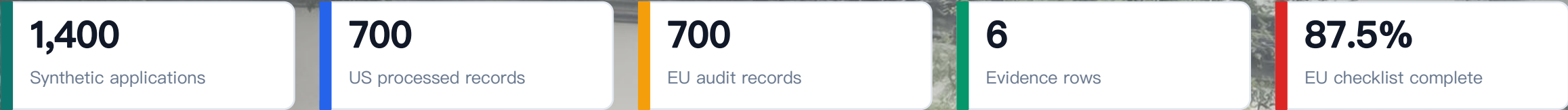
Management takeaway

Approve the tool as a governance monitor — not as a credit engine, legal certification tool or automated bias–remediation system.

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1. Problem & scope: one credit workflow, different regulatory choices

Why a memo or spreadsheet is not enough for cross-jurisdiction AI credit governance.



Regulatory divergence becomes product design

Dimension	United States	European Union
Primary question	Can the creditor provide specific, accurate adverse-action reasons?	Can high-risk AI governance be evidenced before and after deployment?
Sensitive data posture	Hard-drop audit/protected fields from live path.	Use audit-only fields only in segregated testing path.
Tool implication	Reason-code quality + proxy-risk review.	Pre-deployment gate + bias/drift monitoring.

Research boundary

The project does not claim Revolut uses one global credit model or real customer data. The synthetic prototype demonstrates governance mechanics.

RegTech boundary

The tool is valuable because jurisdiction changes data access, computation logic, evidence output and accountable owner — not just slide wording.

2. How the tool works: shared schema, jurisdiction-bound modules

The jurisdiction choice changes data contracts, allowed fields, computation modules and review owners.

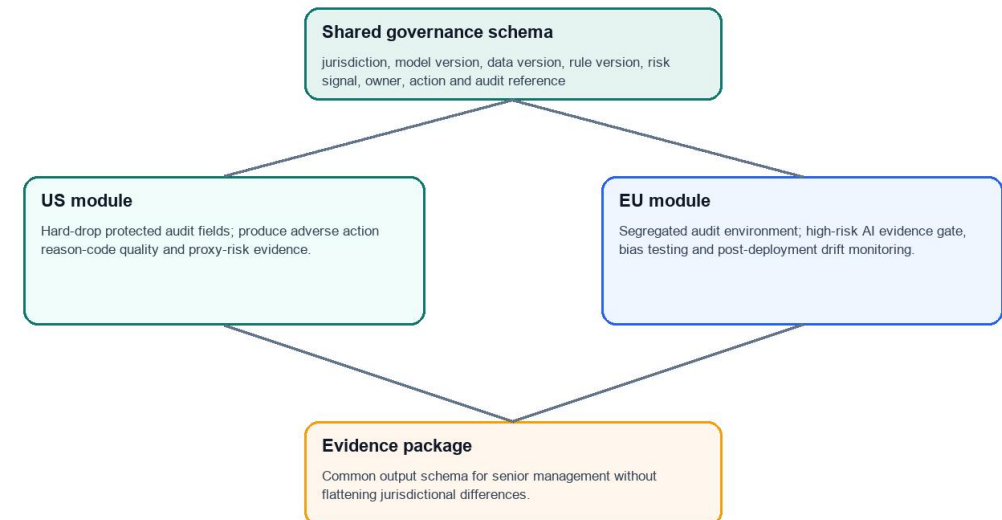
Architecture logic

Shared governance schema	jurisdiction, model version, rule version, risk signal, owner, action
US hard-drop path	removes age / gender_audit / age_band / protected_group_flag / audit_group
EU segregated audit path	retains audit-only fields for controlled bias testing, not live decisions
Rule configuration	owner, legal basis, computation module, version, effective date, rollback
Risk-tiered oversight	Level 0 monitoring → Level 4 emergency suspension

Why this is jurisdiction-aware

A US setting triggers reason-code and proxy-risk outputs. An EU setting triggers high-risk evidence, audit bias testing and drift monitoring. Same evidence schema; different legal mechanics.

Evidence architecture: shared governance, jurisdiction-bound computation



Why this is not decorative

Rules are versioned and owned. A configuration change becomes a governed event, not an invisible code edit.

3. Quantitative prototype: what the tool found in synthetic data

The numbers are synthetic; the governance routing and evidence logic are the deliverable.

25.9%

US low-confidence reasons

31.8%

Postcode denial gap

26.5%

Thin-file denial gap

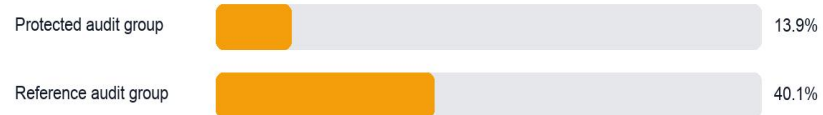
0.35

EU approval ratio

26.5%

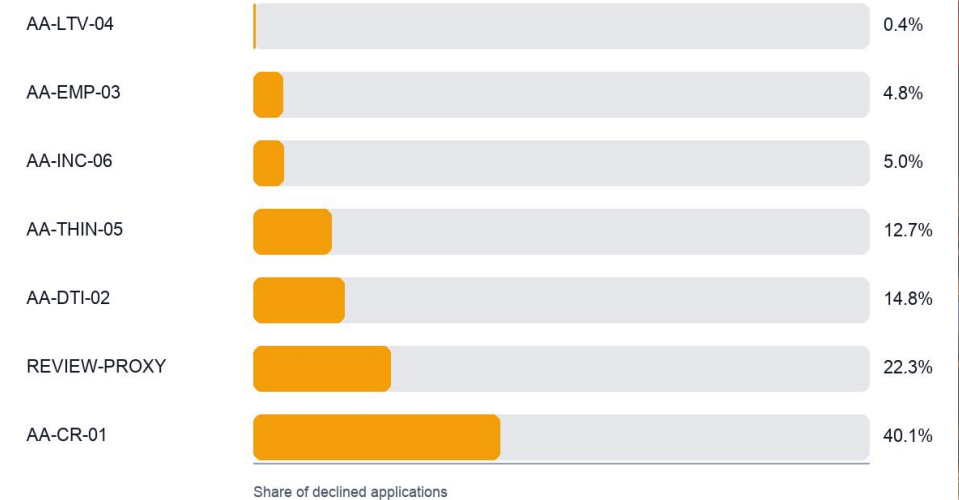
EU FNR gap

EU audit approval rate by group



Approval rate

US reason-code distribution among declined applications

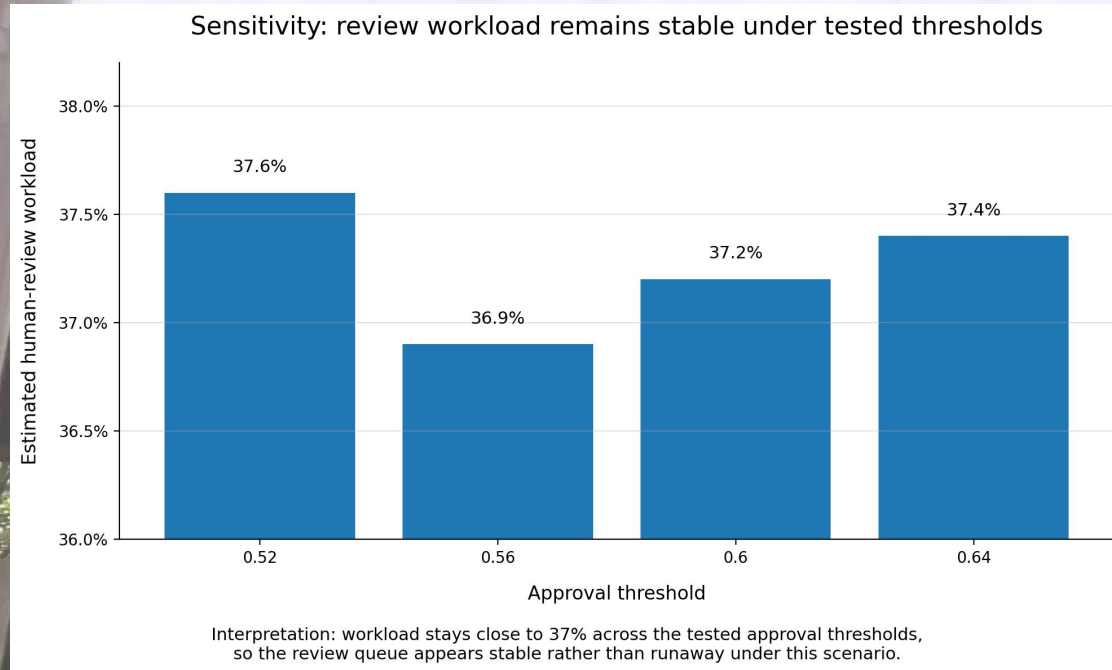


Interpretation

US signals do not prove unlawful discrimination; they trigger fair-lending and reason-code review. EU signals show whether high-risk AI governance evidence is complete before/after deployment.

4. Conclusion: evidence for judgement, not automation of judgement

What management should approve — and what the tool deliberately refuses to do.



Workload conclusion

Review workload stays close to 37% under tested thresholds. The design should use risk-tiered oversight, not broad manual blocking.

Final conclusion

What I did — Designed a jurisdiction-aware governance monitor with synthetic data, config rules, evidence outputs and a senior-management deck.

How it works — US = reason-code quality + proxy-risk review; EU = high-risk governance gate + controlled bias testing + drift monitoring.

What it means — The product names evidence gaps, owners and escalation levels while preserving human judgement.

Strategic boundaries

- No unified sensitive-data pool — Avoids cross-jurisdiction contamination.
- No final adverse-action notice — SHAP-style attribution remains internal aid.
- No automatic model repair — Keeps remediation inside model governance.
- No legal certification — Outputs evidence and escalation, not legality.